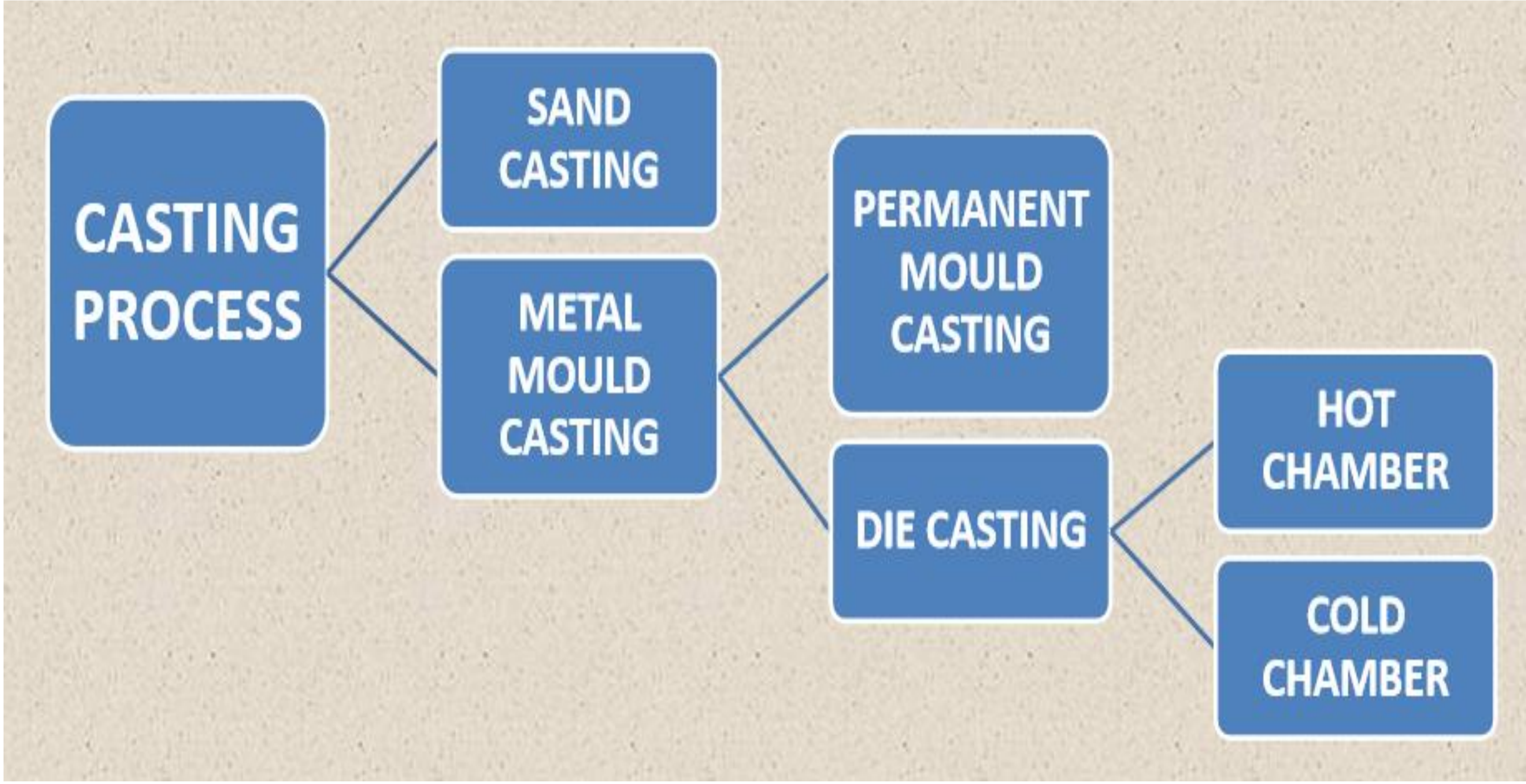
CASTING

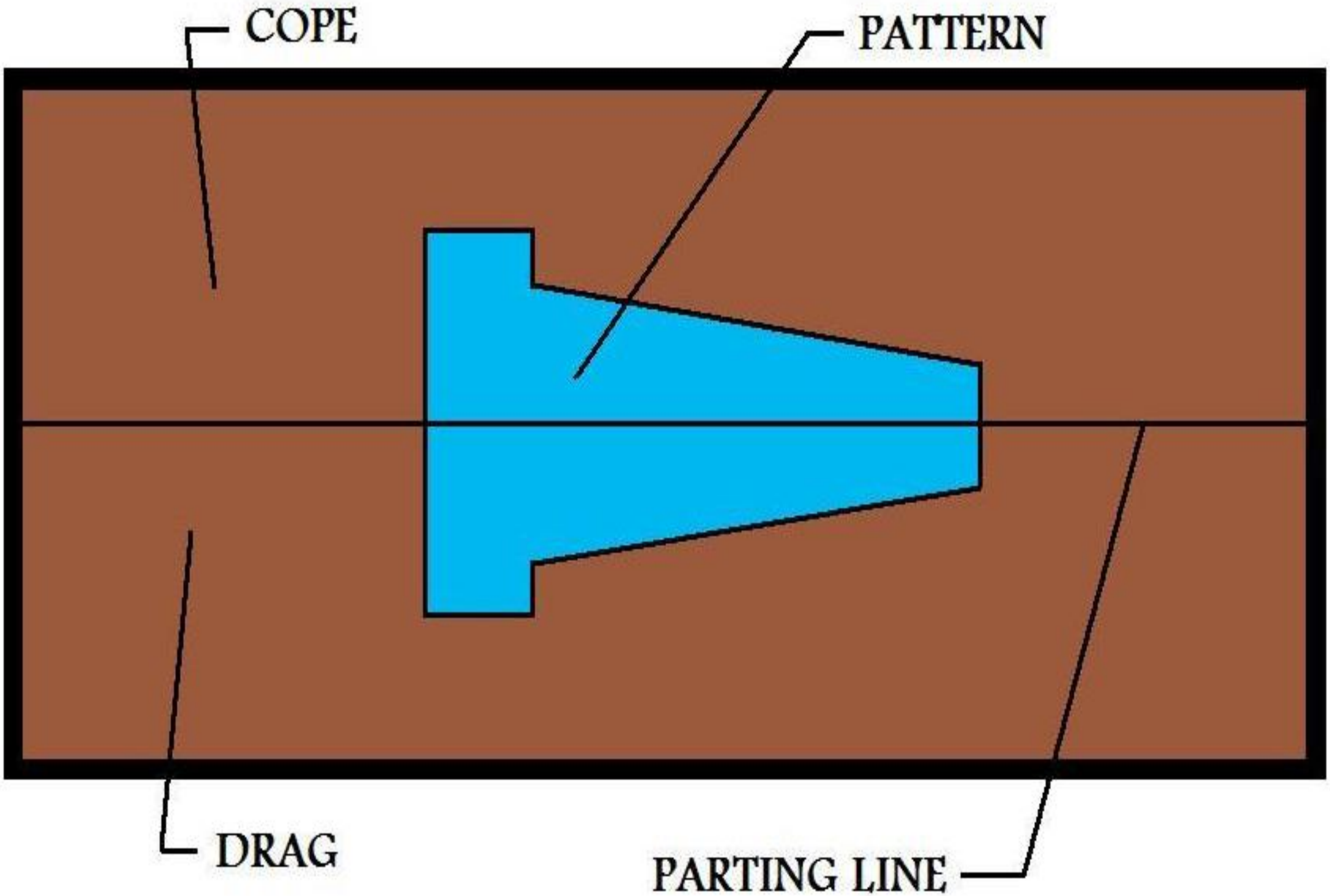
- It is a manufacturing process in which molten metal is poured in a mould or cavity and allowed to solidify
- Molten metal on solidification gets the shape of the mould
- Mould has the shape of the product to be made


CASTING PROCESS
SAND CASTING
METAL MOULD CASTING
PERMANENT MOULD CASTING
DIE CASTING
HOT CHAMBER
COLD CHAMBER

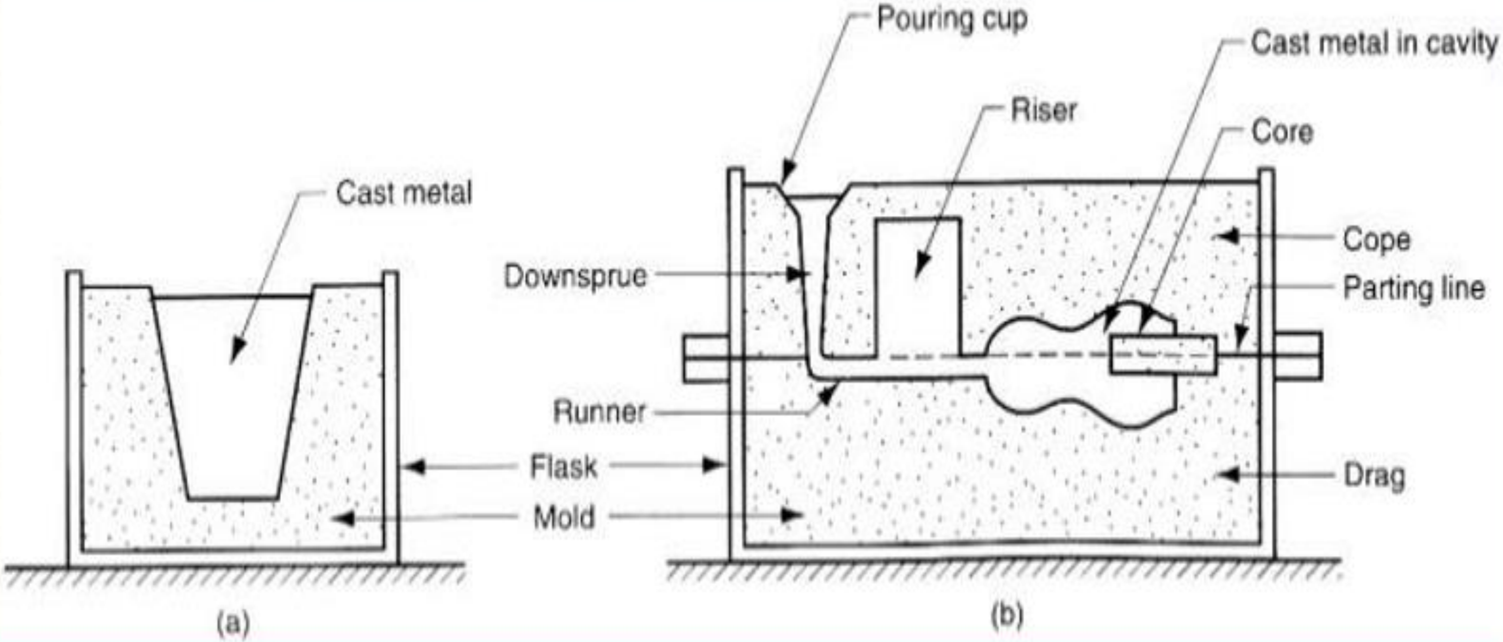
SAND CASTING

- Molten metal: metal in the form of a liquid
- Mould: negative print of the product to be cast (cavity whose geometry determines the shape of the cast part), open and closed mould
- Moulding: process of making mould of desired shape using sand, pattern and core
- Mould consists of two halves a) Cope: upper half of the mould b) Drag: bottom half of the mould
- *Pattern: model or replica of the component to be made by casting(mould forming tool)*

Molten metal


COPE
PATTERN
DRAG
PARTING LINE

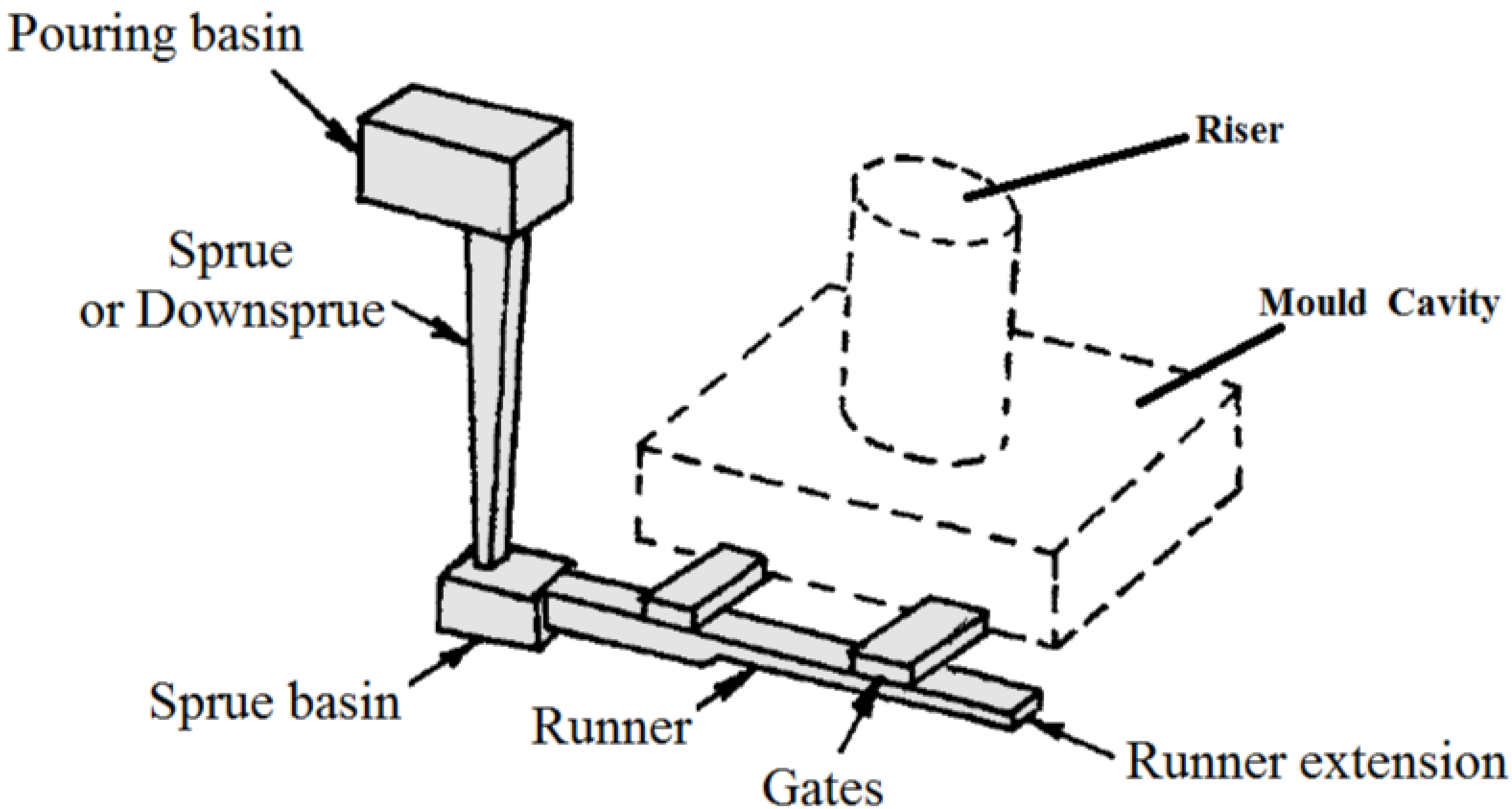
Open and Closed Mould



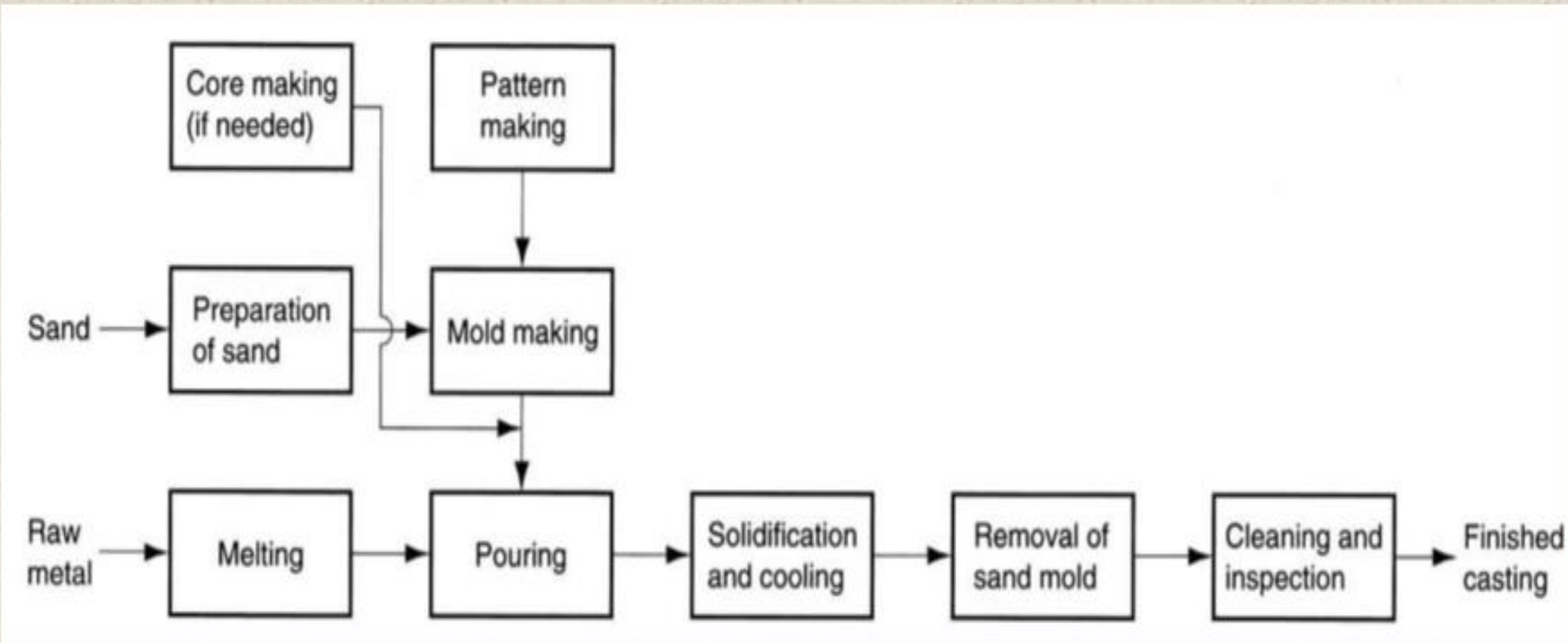
PARTS OF GATING SYSTEM

- Pouring basin: top of the mould for pouring the molten metal at the required rate into the mould cavity
- Sprue: vertical passage made through the cope for connecting pouring basin with the gate
- Runner: for connecting the sprue and gate
- Gates: passage for connecting the base of the runner with the mould cavity
- Riser: passage made in the cope to permit molten metal to rise up after filling the mould cavity

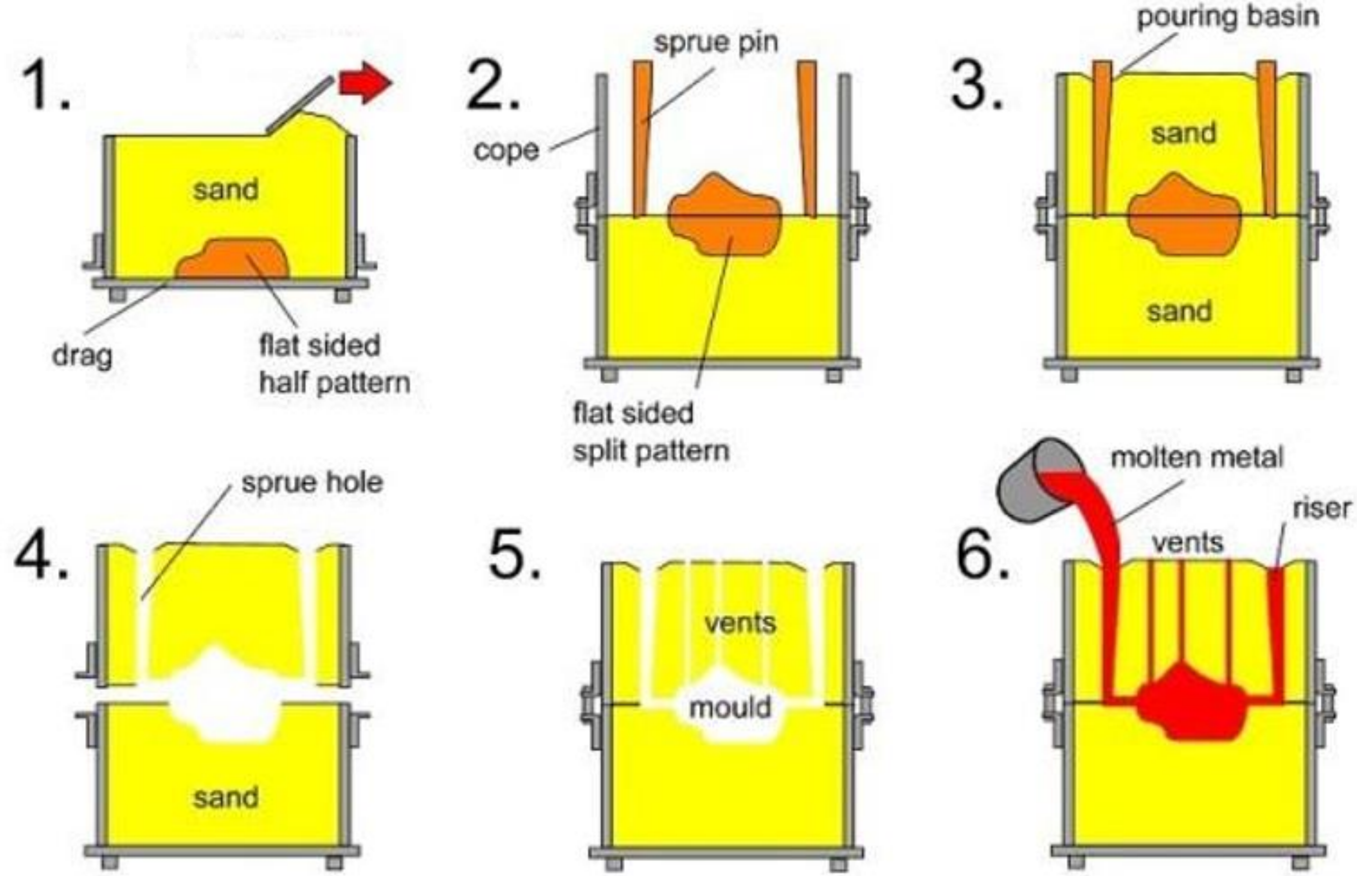
GATING SYSTEM



STEPS IN SAND CASTING



SAND CASTING PROCEDURE



MOULDING SAND PROPERTIES

- FLOWABILITY: behave like a fluid, sand to get compacted to uniform density
- GREEN STRENGTH: strength of the sand in moist condition
- DRY STRENGTH: strength of the sand in dry condition
- POROSITY/ PERMEABILITY: ability to allow the passage of mould gases
- REFRACTORINESS: ability of the sand to withstand high temperature

- ADHESIVENESS: ability of the sand to stick on to the mould walls
- CHEMICAL STABILITY: resist chemical reaction
- COLLAPSIBILITY: ability of the sand to collapse after the casting solidifies
- FINENESS: ability of the sand to produce smooth surfaced castings
- COEFFICIENT OF EXPANSION: less coefficient of expansion
- DURABILITY: ability of the sand to be used again and again

- Advantages
- Production process is simple
- Cost of casting is low
- Sand can be reused

- Disadvantages
- More chances of defects
- Does not impart good surface finish

CASTING DEFECTS

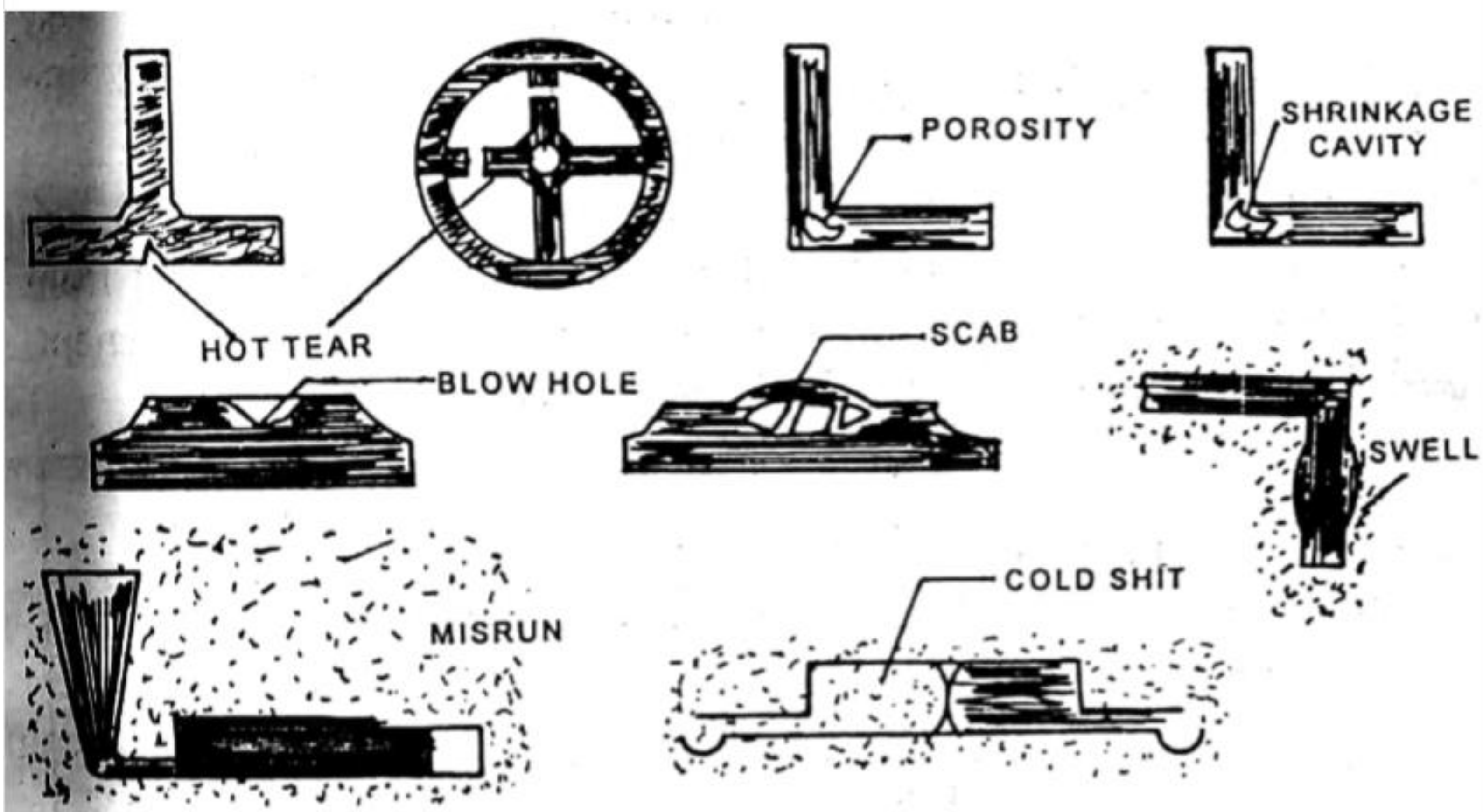


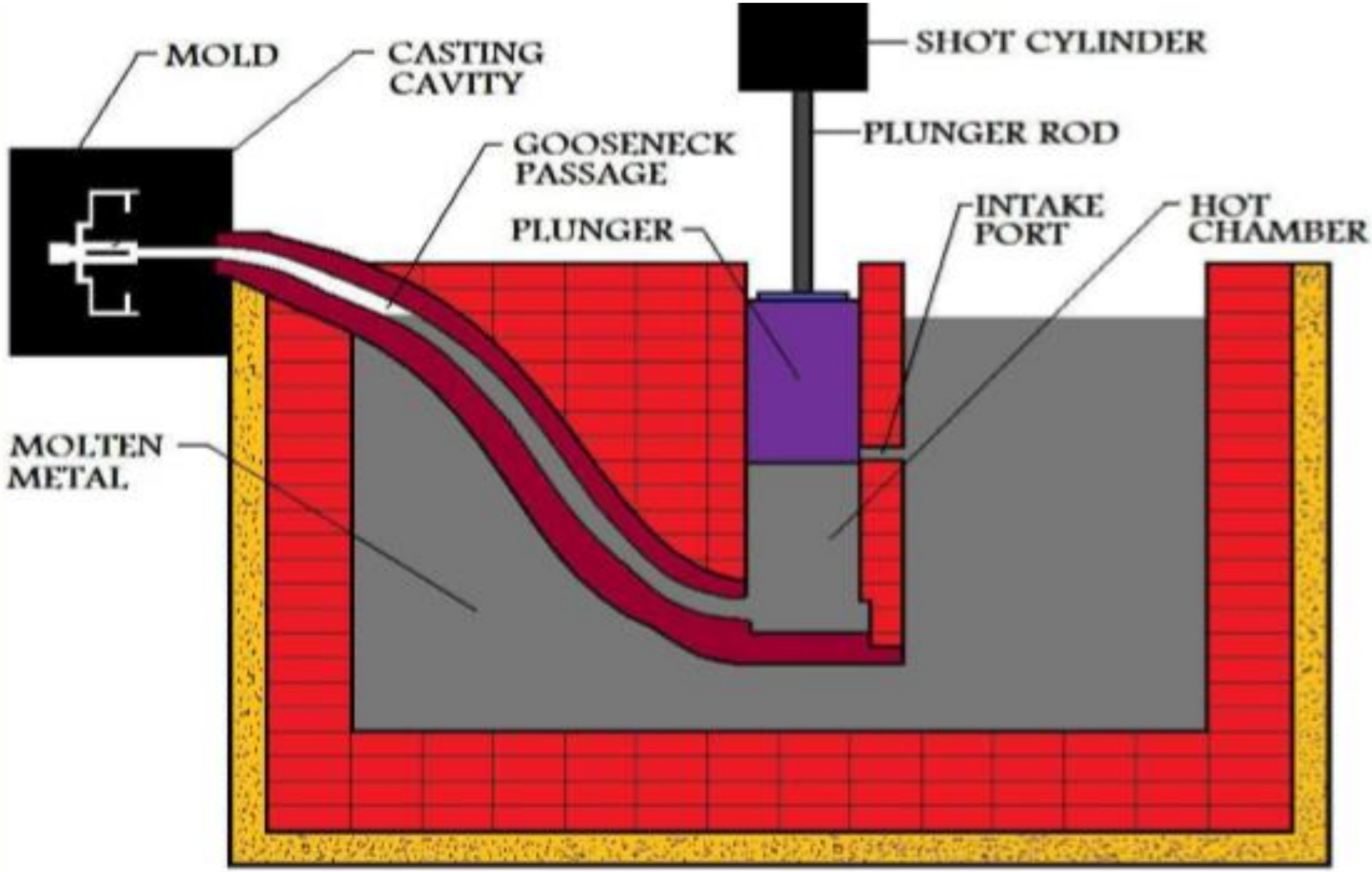
Fig. 3.7 COMMON DEFECTS IN CASTINGS

CASTING DEFECTS

- Blow holes- spherical voids formed due to evolution of entrapped gases in the metal, vapourisation of moisture in sand etc
- Cold shut and misrun – are discontinuities in the casting as a result of poor fluidity of the molten metal. Misrun are formed when the entire section is not filled during pouring before solidification.

Cold shuts formed when two streams of metal do not fuse together.

- Hot tear – are cracks in the casting as a result of contraction stresses after solidification
- Mismatch is a shift of the individual parts of a casting with respect to each other. Mismathing of cope and drag.
- Shrinkage cavities – are voids in the casting due to insufficient feeding, poor casting design, high temp of pouring metal etc.
- Swell – is an expansion of mould cavity by metal pressure. It is due to insufficient ramming and too rapid pouring of metal
- Scabs – are lumps of excess metal on the casting as a result of erosion of mould by the stream of molten metal.



HOT CHAMBER DIE CASTING

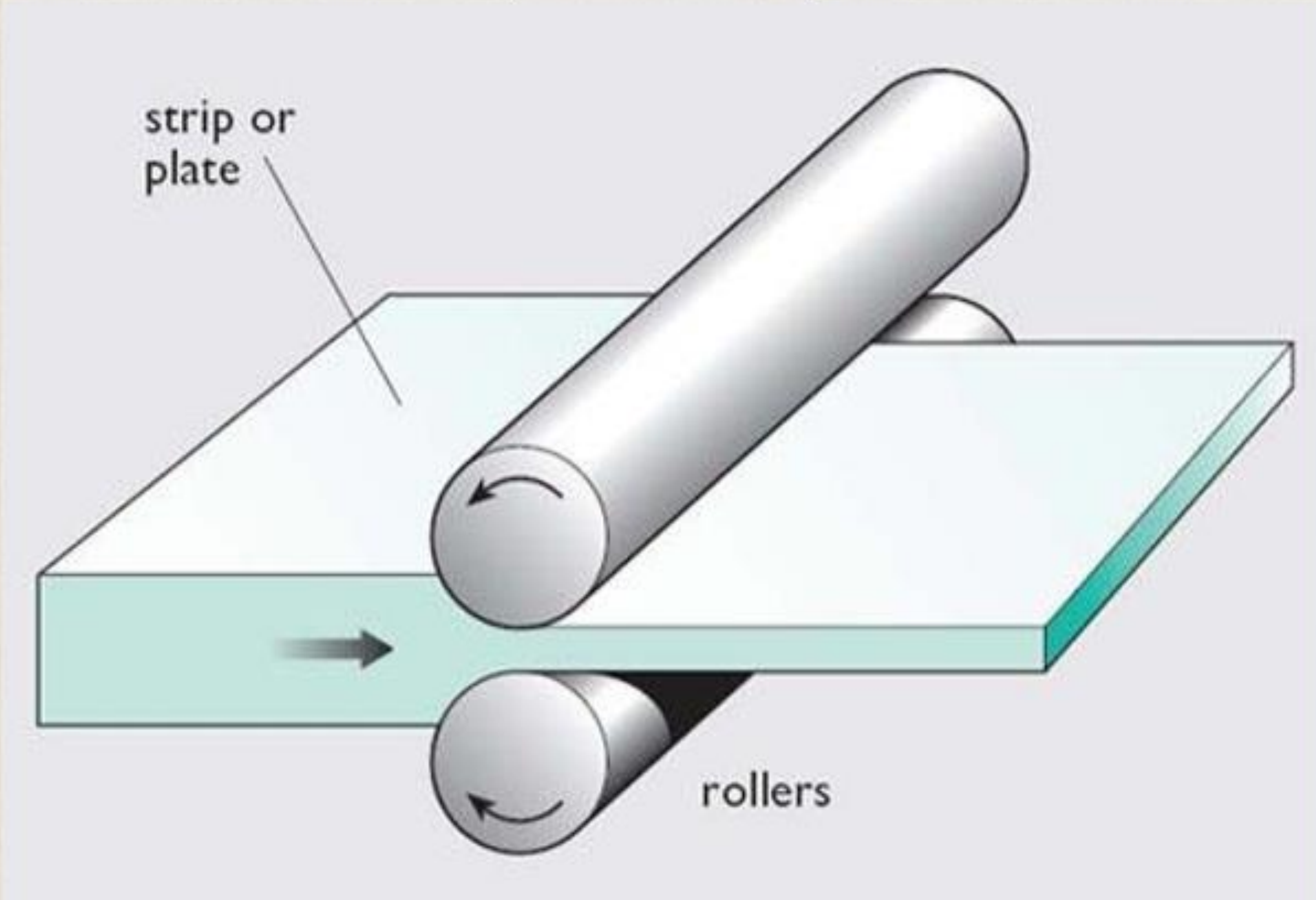
Metal Forming By Deformation

- Process in which shape of the metals are changed to desire shapes by subjecting them to stresses greater than yield stress of the metal • It is a deformation process
- Types are
- Rolling
- Forging
- Extrusion

ROLLING

- Process of plastically deforming metal by passing it between rolls
- Cylindrical rolls are used to reduce the cross sectional area of a bar or plate with a corresponding increase in the length
- Process of rolling basically consists of passing metal between two rolls rotating in opposite direction at the same speed
- Two types
- Hot rolling
- Cold rolling

1. Rolling



1. Hot Rolling

- Process in which metal is fed to the rolls after being heated above the recrystallization temperature

2. Cold Rolling

- In cold rolling, metal is fed to the rolls when it is below the recrystallization temperature

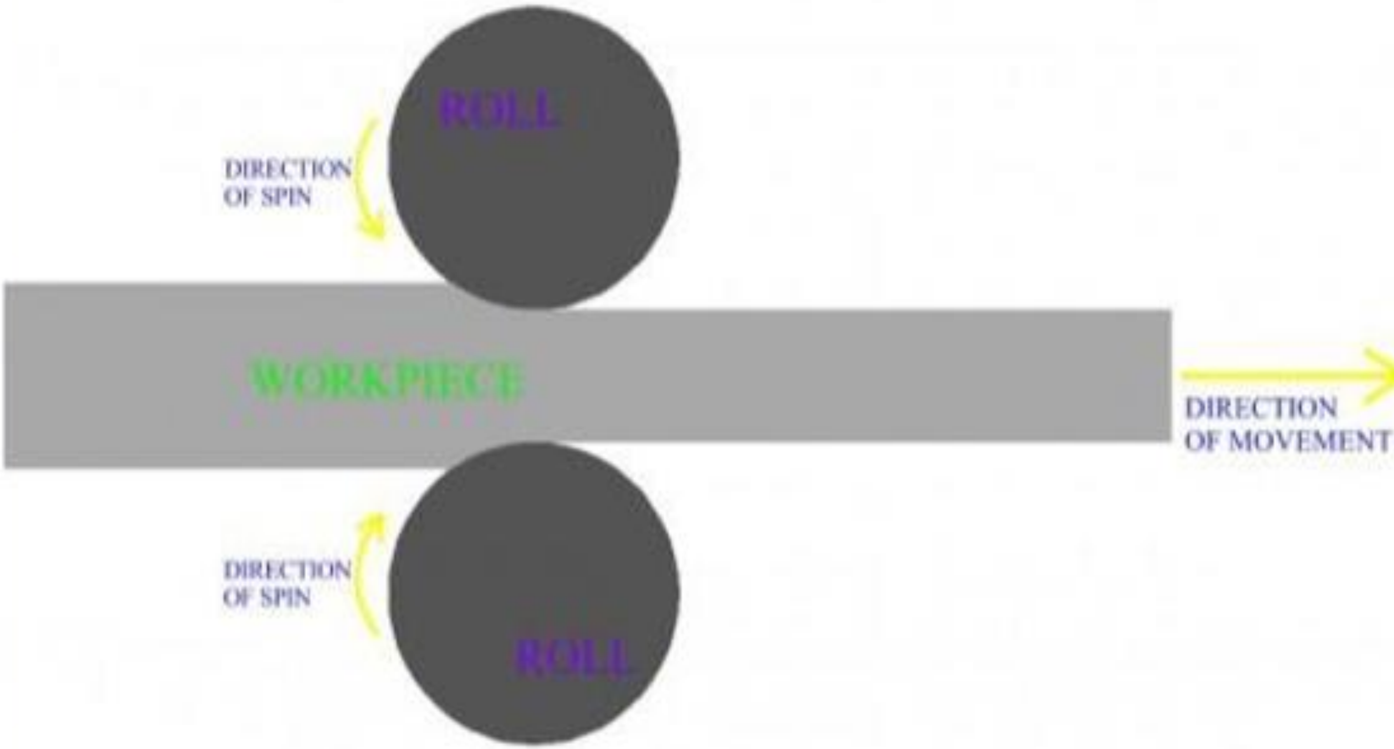
Types of rolling mills

- a) Two high mill
- b) Three high mill
- c) Four high mill
- d) Cluster mill
- e) Tandem mill

TWO HIGH MILL

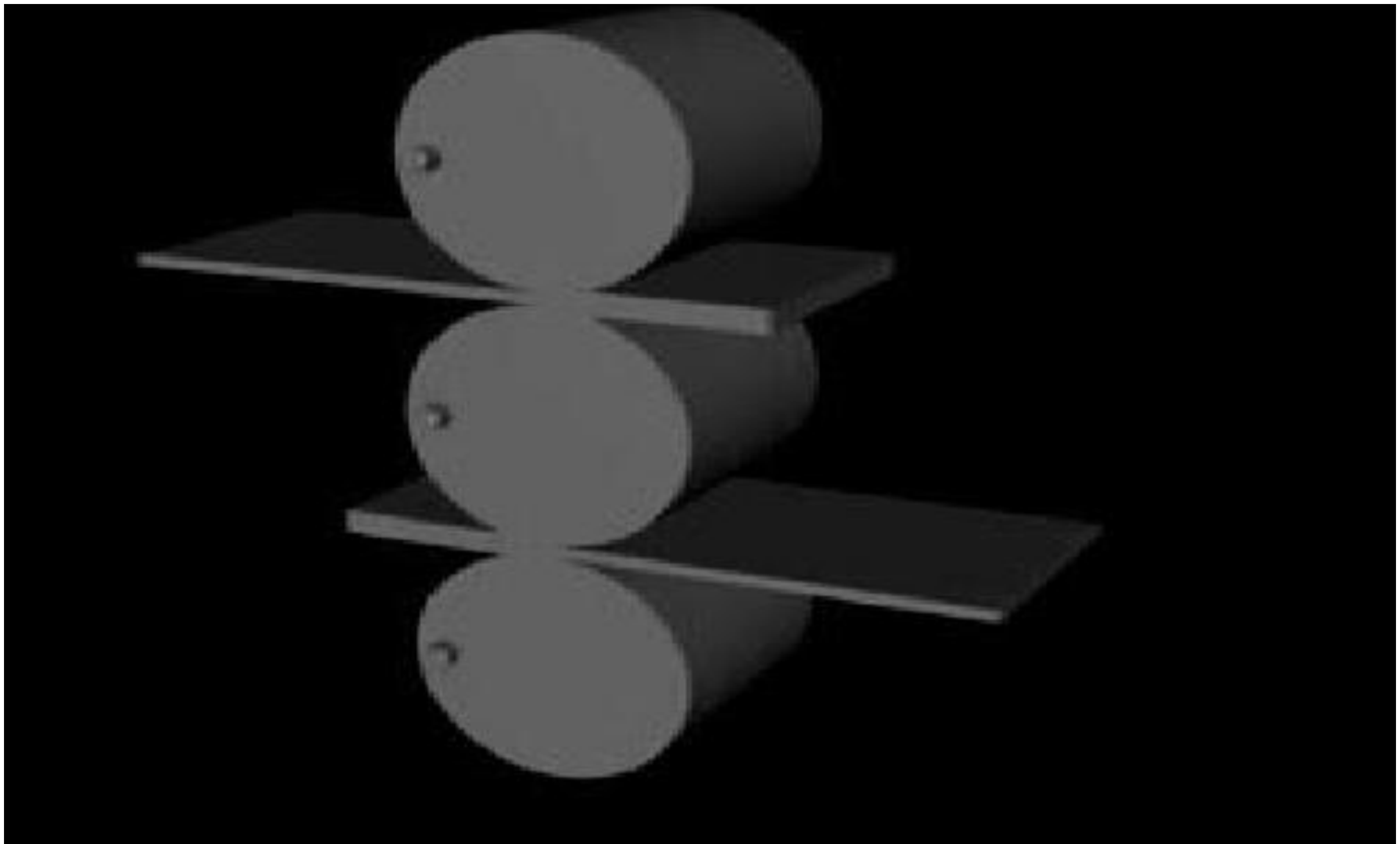
- Two rolls
- Lower roll will be fixed
- Upper roll can be moved to adjust the space between the rolls
- Both the rolls rotate at the same speed but in opposite directions

TWO HIGH ROLLING MILL



THREE HIGH ROLL MILL

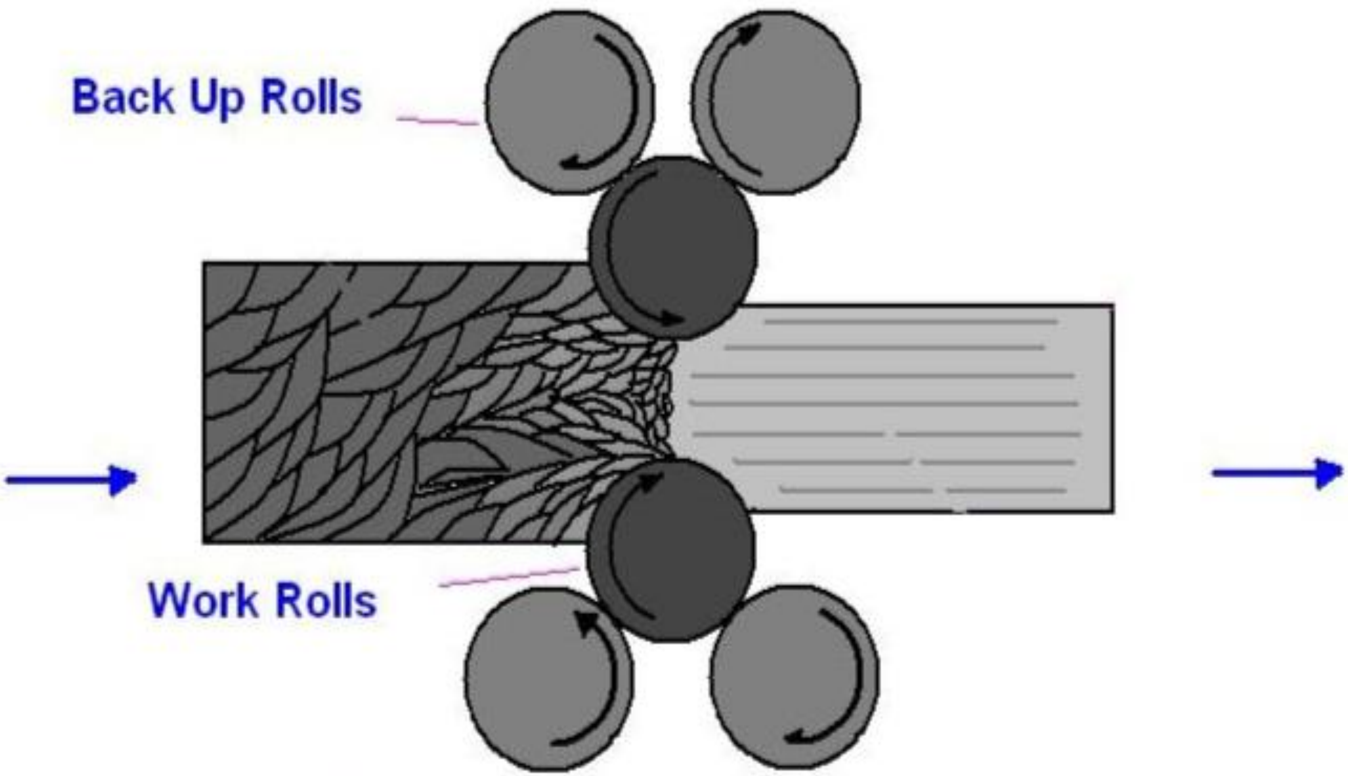
- Three rolls positioned one over another
- Upper and lower rolls rotate in the same direction
- Middle roll rotates in the opposite direction
- Middle roll is fixed
- Upper and lower rolls are moved to adjust the roll gap



Three-high Rolling mill

Cluster Mill

- Used for rolling very thin sheet or foils
- It consists of a Pair of working rolls of very small diameter, supported by a number of back up rolls on either side



Cluster Rolling Mills

FORGING

- Process of changing the shape of metals when it is in the plastic state, by applying compressive force

- Hot forging – forging at high temperature
- Cold forging- forging at room temperature

- Forged product has better mechanical properties than a cast one

Components produced by forging

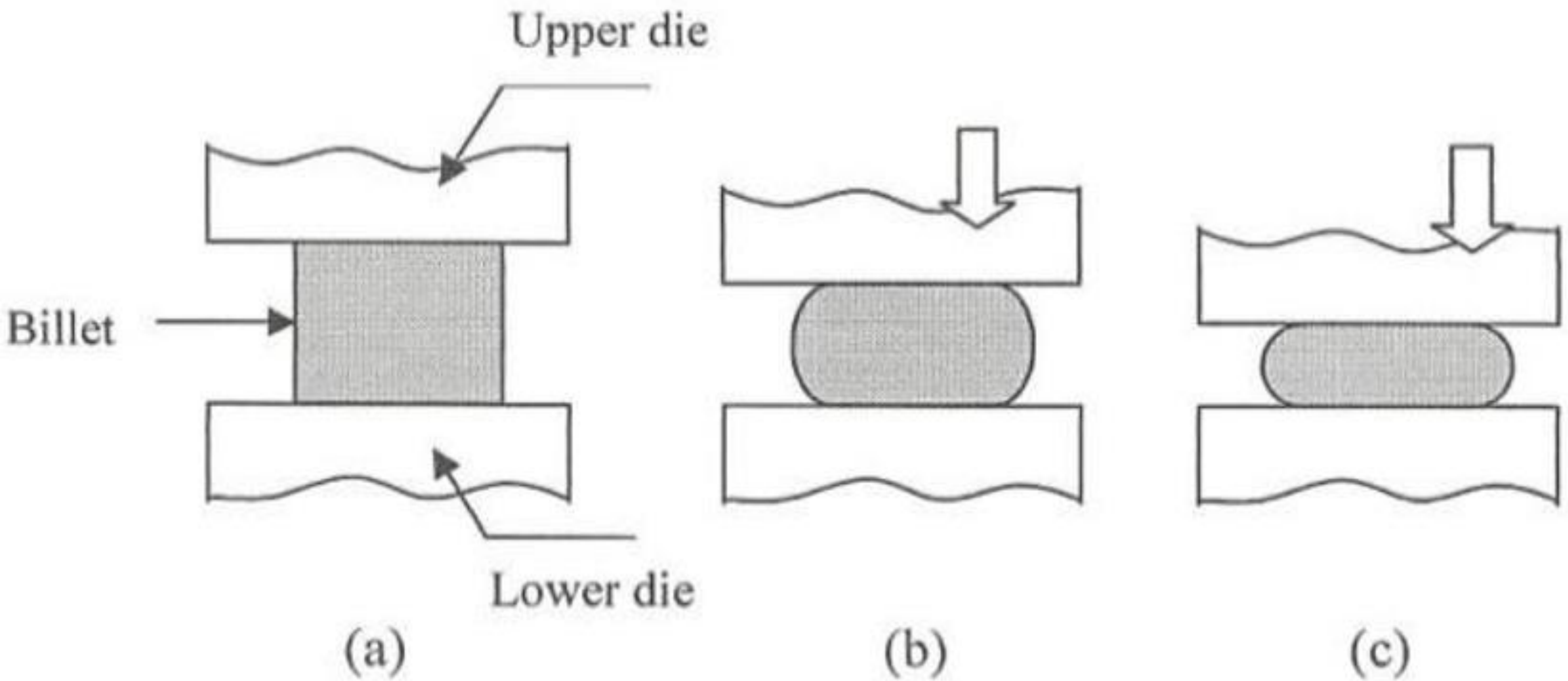
- Nails
- Bolts
- Spanners
- Crane hooks
- Axles
- Crankshafts
- Connecting rods

Dies used for forging

- a) Open die forging
- b) Closed / Impression die forging

Open Die Forging

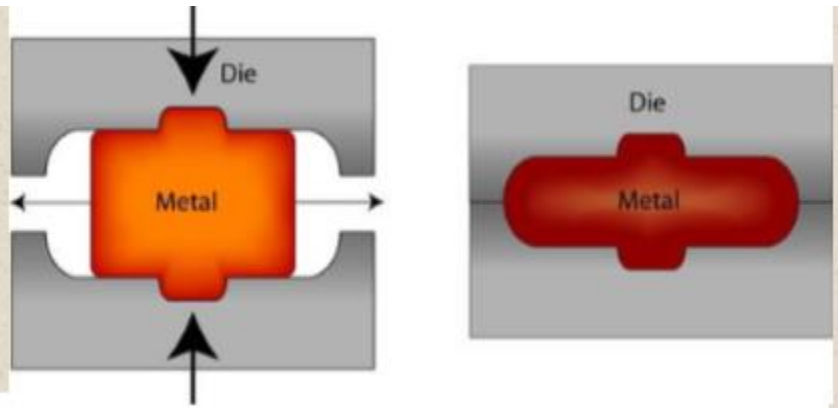
- Work piece is upset, compressed or forged between two flat dies
- Used for simple shapes and low production volumes

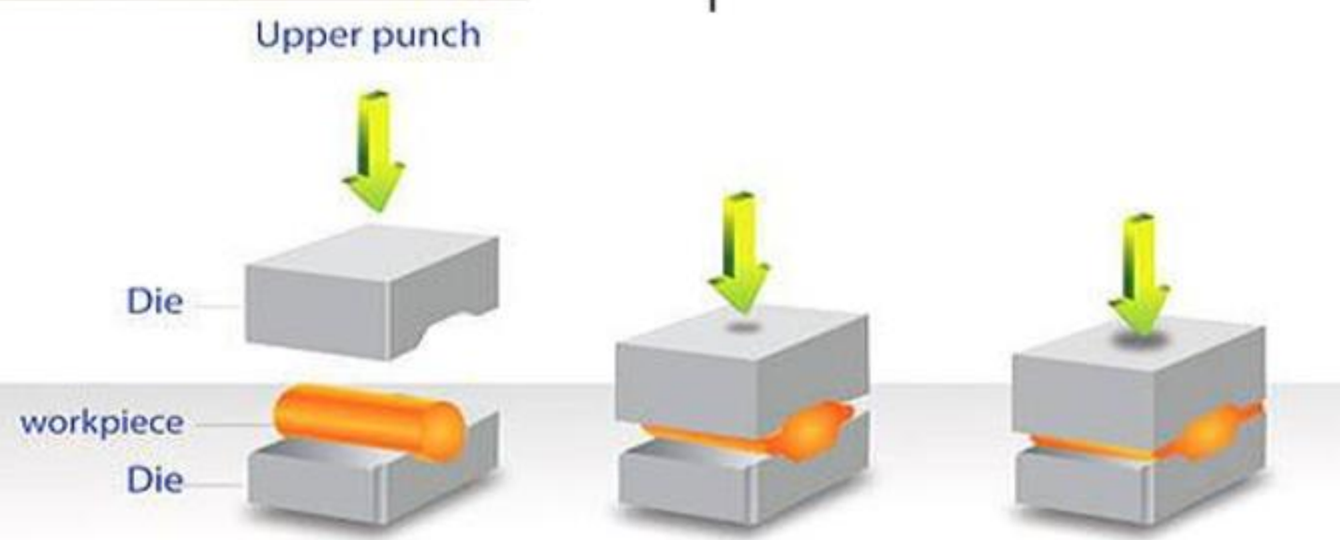


Upper die
Billet
Lower die
(a)
(b)
(c)

b) Closed/Impression Die Forging

- Work piece takes the shape of the die cavity while being forged between two shaped dies
- Used for forging complicated shapes
- Process is usually carried out at elevated temperatures


Die
Metal
Die
Metal


Upper punch
Die
workpiece
Die

Types of Forging

- A. Hand forging
- B. Drop forging
- C. Press forging

A. HAND FORGING

- Traditional forging operation carried out by blacksmith in a section of workshop called smithy
- Hand tools are used for forging(eg. Hammer, chisel…etc)
- Not suitable for mass production

Hand forging operations

- a) Upsetting
- b) Drawing down(necking down)
- c) Setting down
- d) Bending
- e) Welding
- f) Cutting
- g) Swaging
- h) Drifting
- i) Fullering

a. Upsetting

- Process of increasing the cross sectional area of a bar at any desired portion at the expense of length of the bar
- Portion to be upset is heated and then hammered axially

b. Drawing Down

- Process of reducing the cross section of a bar by increasing its length

c. Setting Down

- Local thinning down operation using a set of hammer

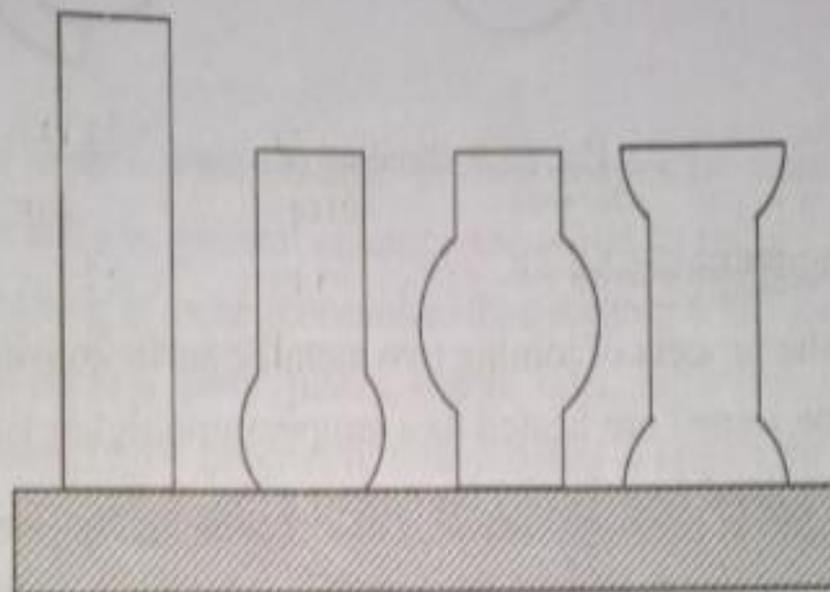
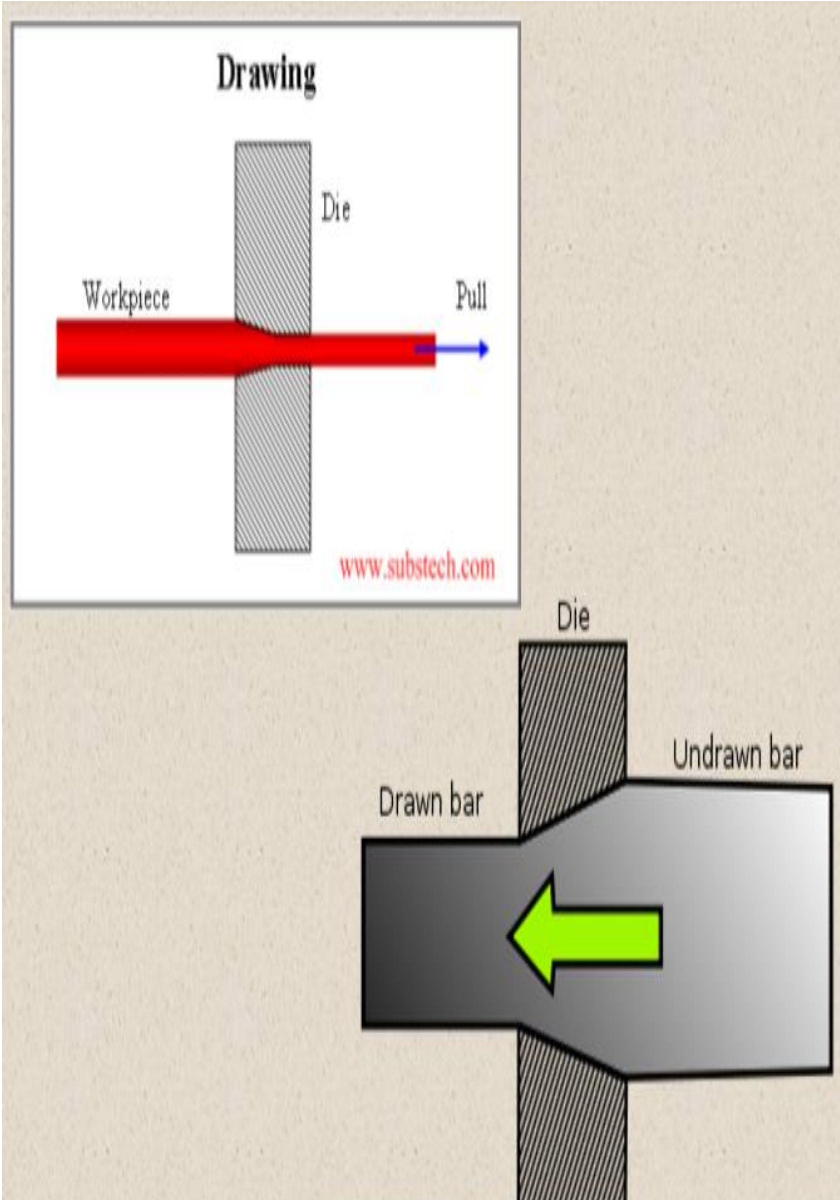
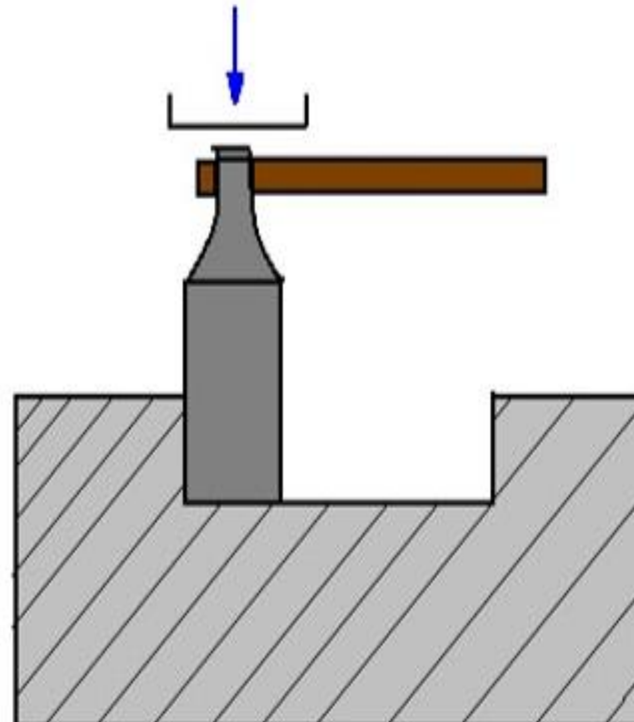


Fig. 5.37. Upsetting

Fig. 5.37 shows the effect on a round billet (a) , when hammering it by healing only at on end (b) , only at the centre (c), and at both ends keeping the centre portion unheated (d).



Setting Down



Set Hammer

d. Bending

- Bars and rods are bent to form rings, hooks…etc

e. Welding

- Joining two metallic surfaces
- Surfaces to be joined are heated to a temperature of about $1000^{o}C$
- Metal surfaces are joined by applying pressure at the mating surfaces

g. Swaging

- Operation by which the required cross sectional shape is obtained
- Two swage blocks, top swage and bottom swage are used for swaging operation
- Work piece is held between the top and bottom swages and is hammered

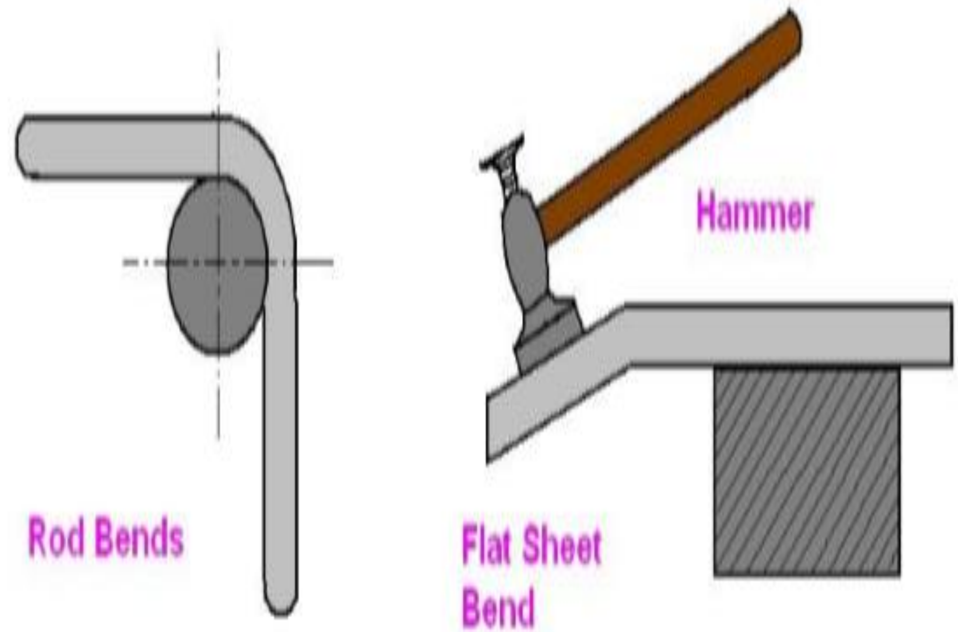
h. Drifting

- Process of increasing the diameter of a punched hole
- Drift which has tapered end is made to pass through the punched hole to produce a finished hole

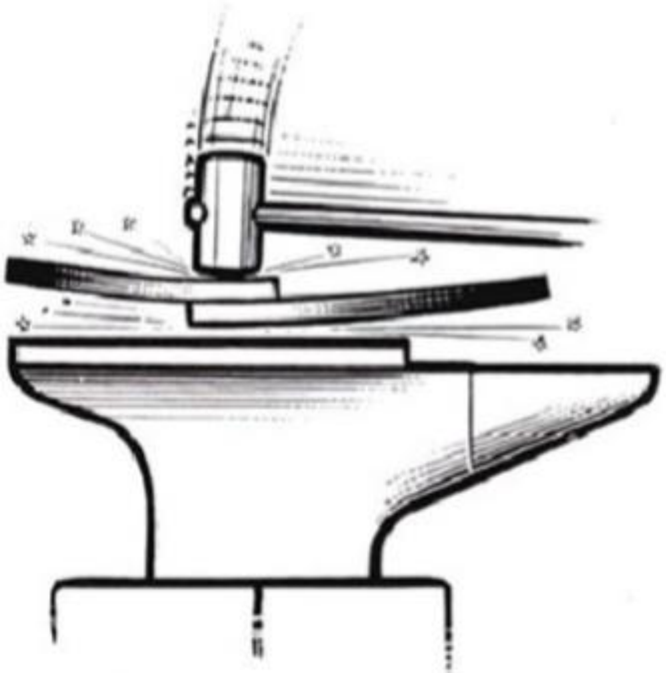
i. Fullering

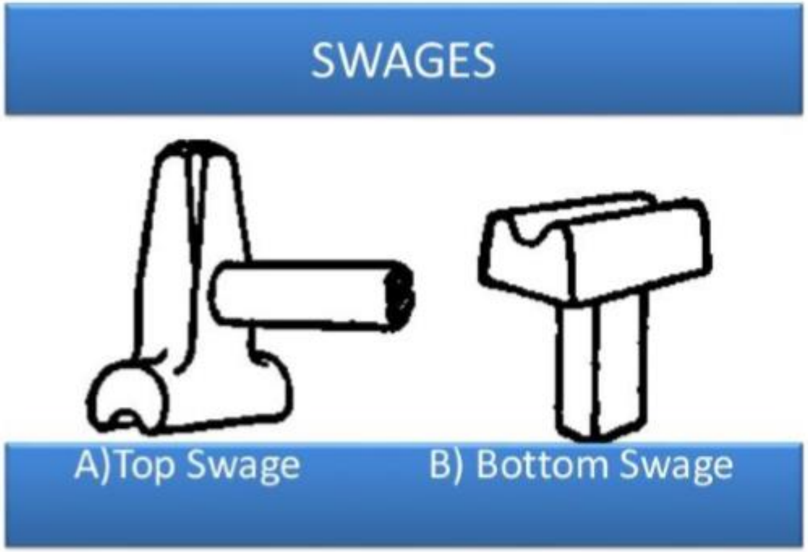
- Reduce the cross sectional area of a portion of a stock
- Metal flow outwards and away from the center of the fullering tool

Bending

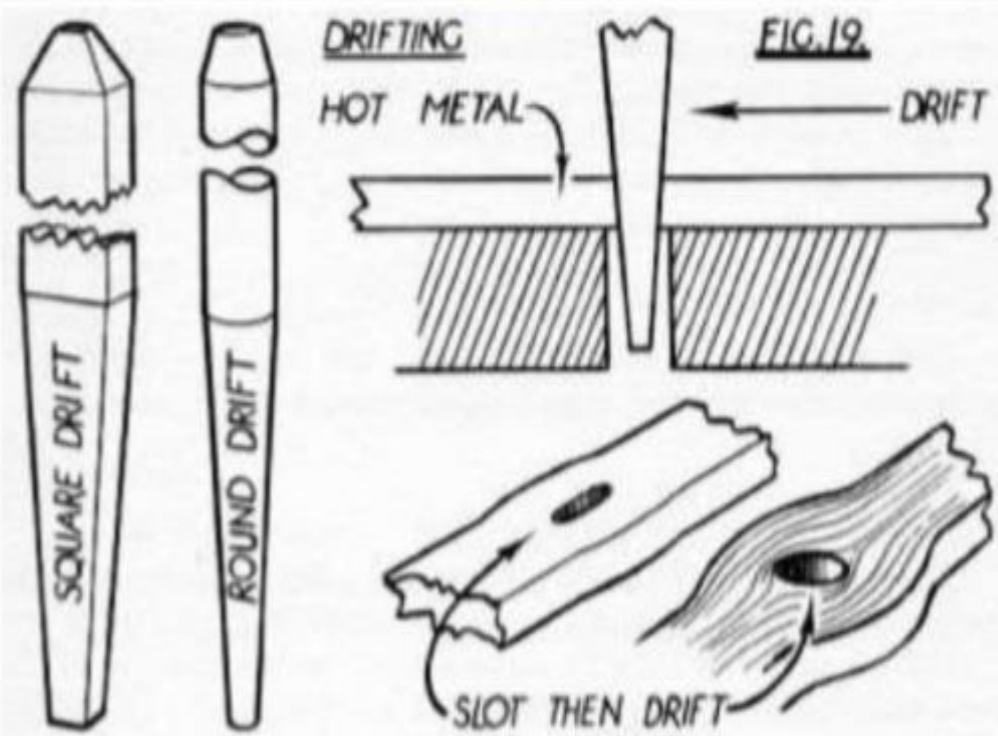


Forge welding





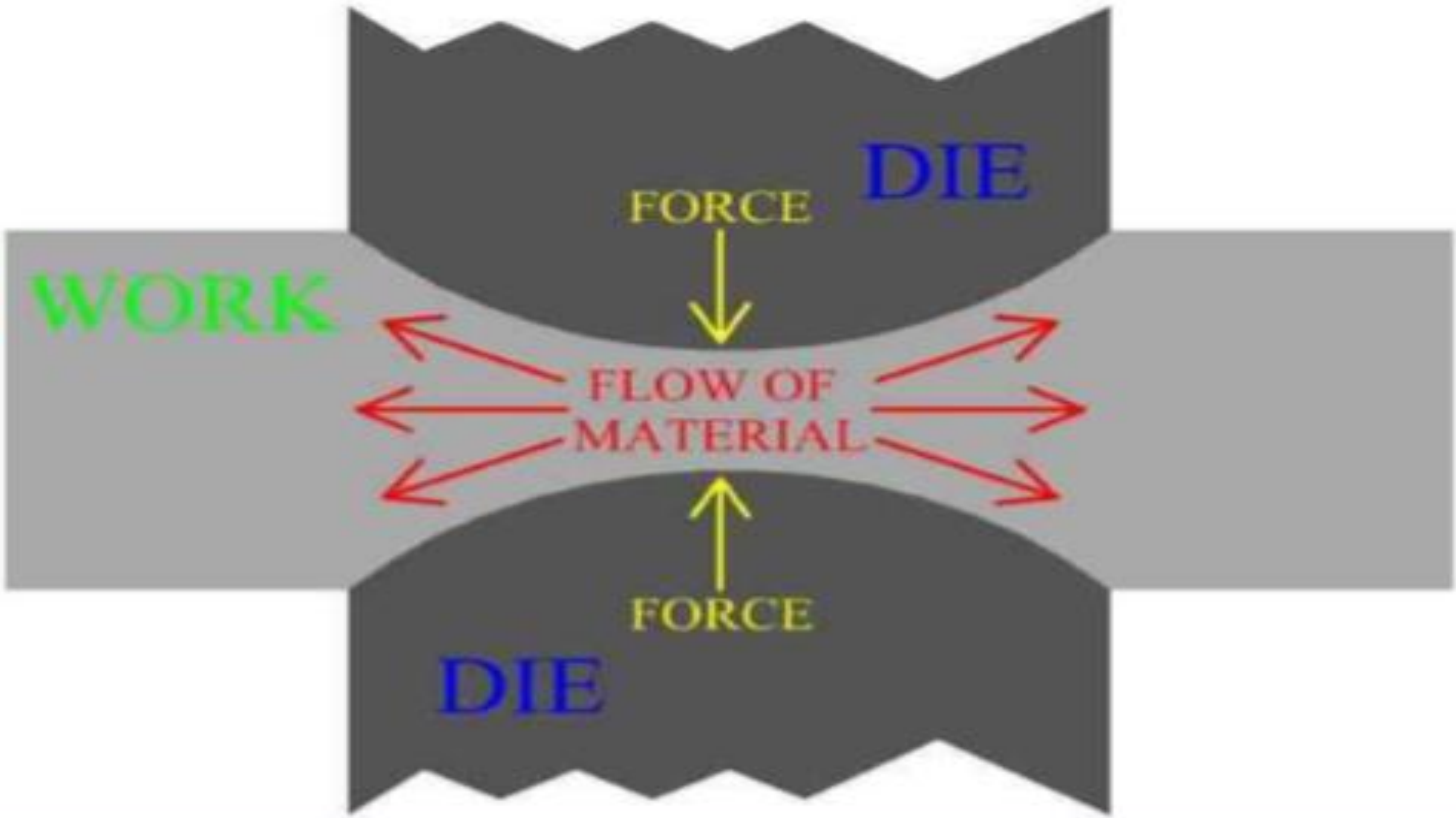
DRIFTS


SQUARE DRIFT
ROUND DRIFT
DRIFTING
FIG. 19.
HOT METAL
DRIFT
SLOT THEN DRIFT


drift

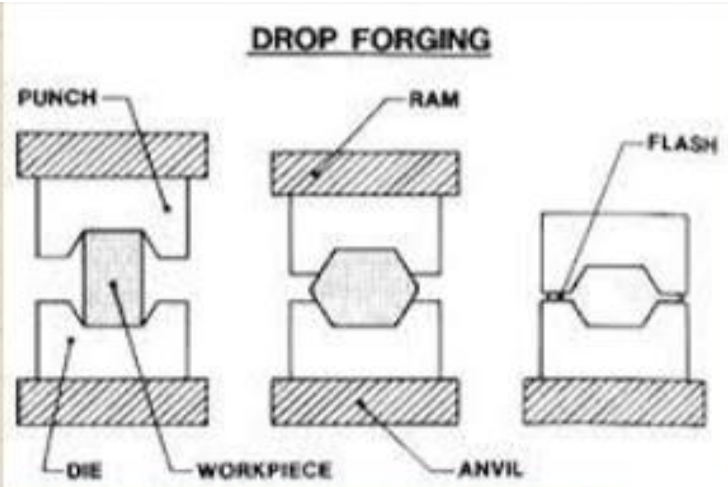

DRIFT

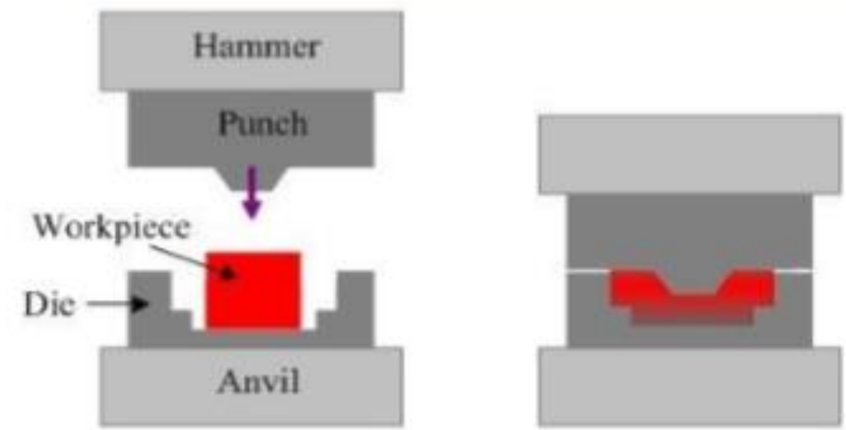
FULLERING



B. DROP FORGING

- Force for shaping the component is applied in a series of blow by using drop hammers
- Open die or closed dies are used for this purpose



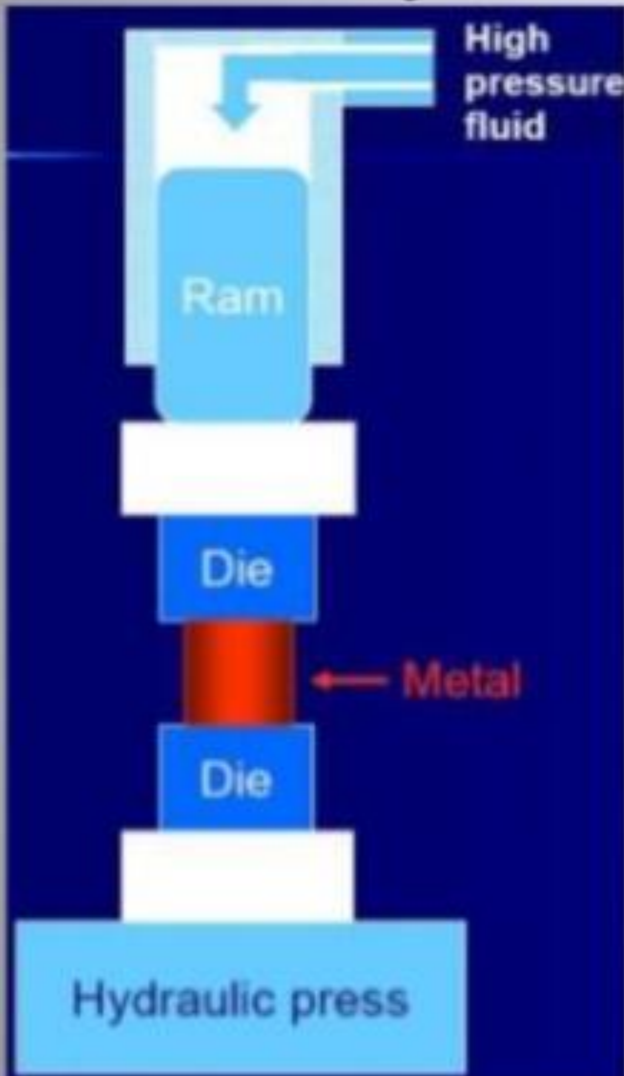


DROP FORGING

C. PRESS FORGING

- Process is similar to drop forging but for the method of application of force
- In this case the force is applied by a continuous squeezing operation by means of a hydraulic press
- Mass production technique

Hydraulic press forging



- Using a hydraulic press or a mechanical press to forge the metal, therefore, gives continuous forming at a slower rate.
- Provide deeper penetration.
- Better properties (more homogeneous).
- Equipment is expensive.

EXTRUSION

- Process of forcing a metal enclosed in a container to flow through the opening of a die
- Metal is subjected to plastic deformation
- Metal undergoes reduction and elongation during extrusion
- Used for manufacture rods, tubes, circular, rectangular, hexagonal and other shapes both in hollow and solid form

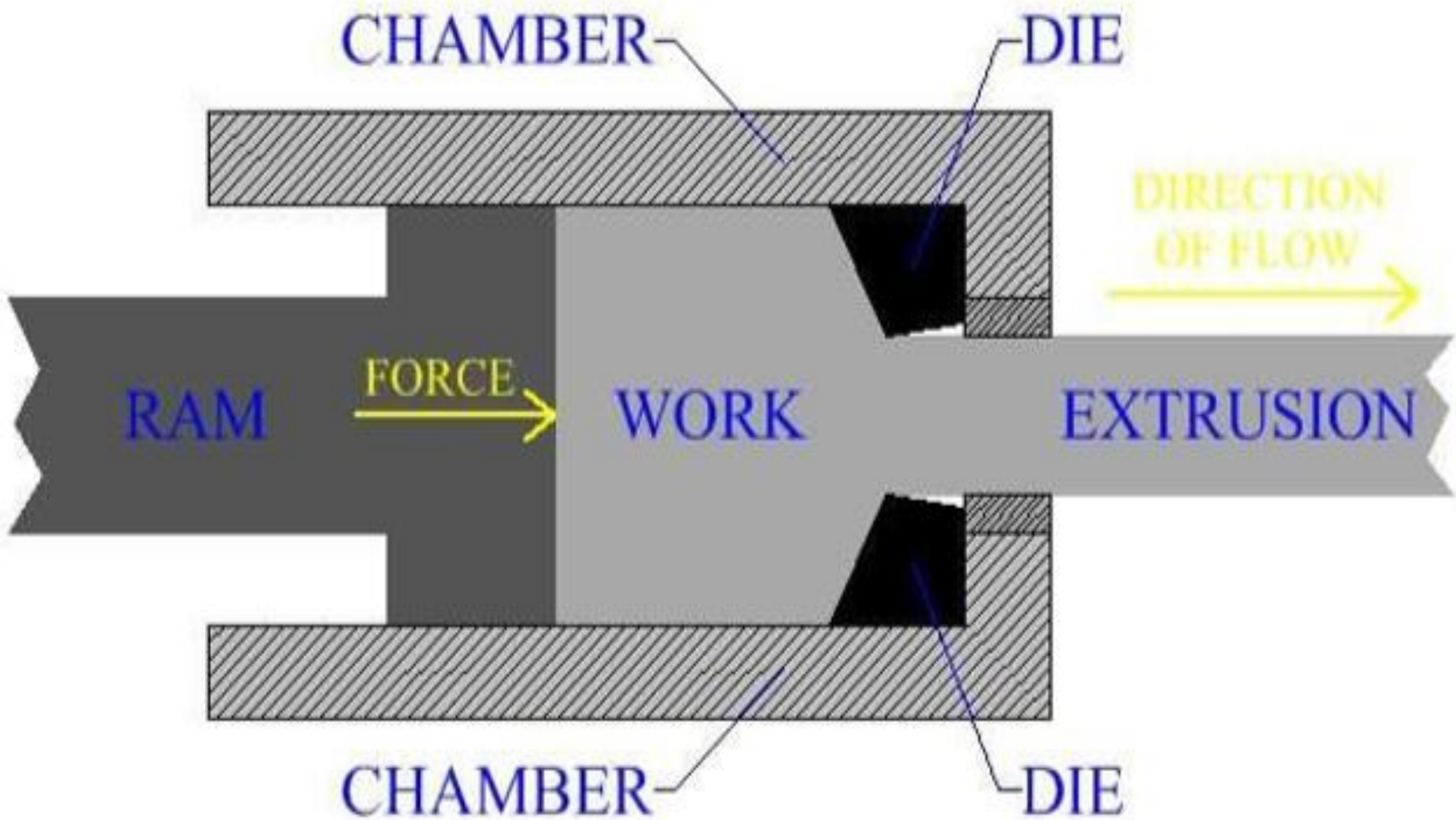
Types of Extrusion

- a) Direct Extrusion
- b) Indirect Extrusion
- c) Cold Extrusion/ Impact Extrusion

a. Direct Extrusion

- Also called forward extrusion
- Flow of metal through the die is in the same direction as the movement of ram
- Hot billet(work piece) is used

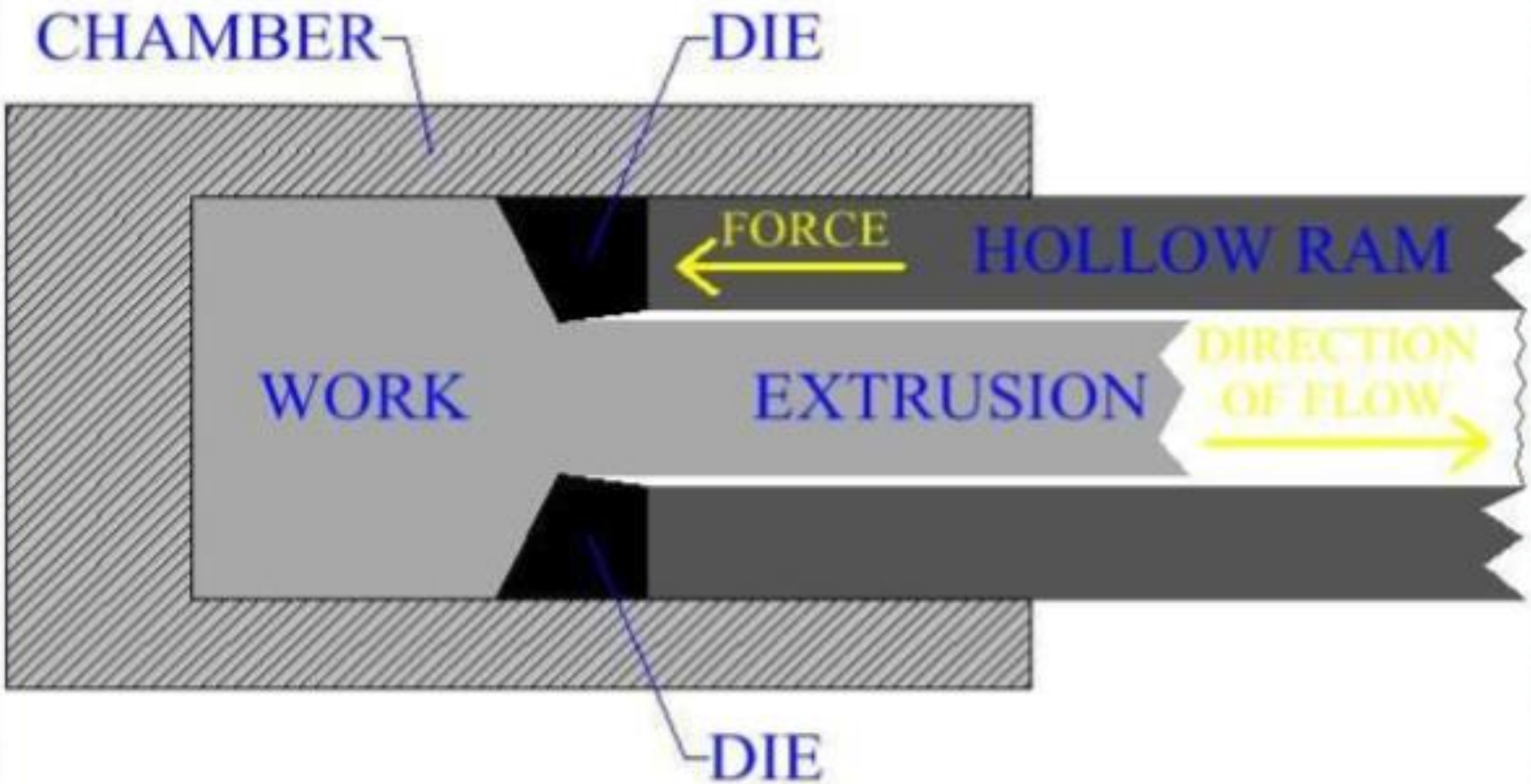
DIRECT EXTRUSION

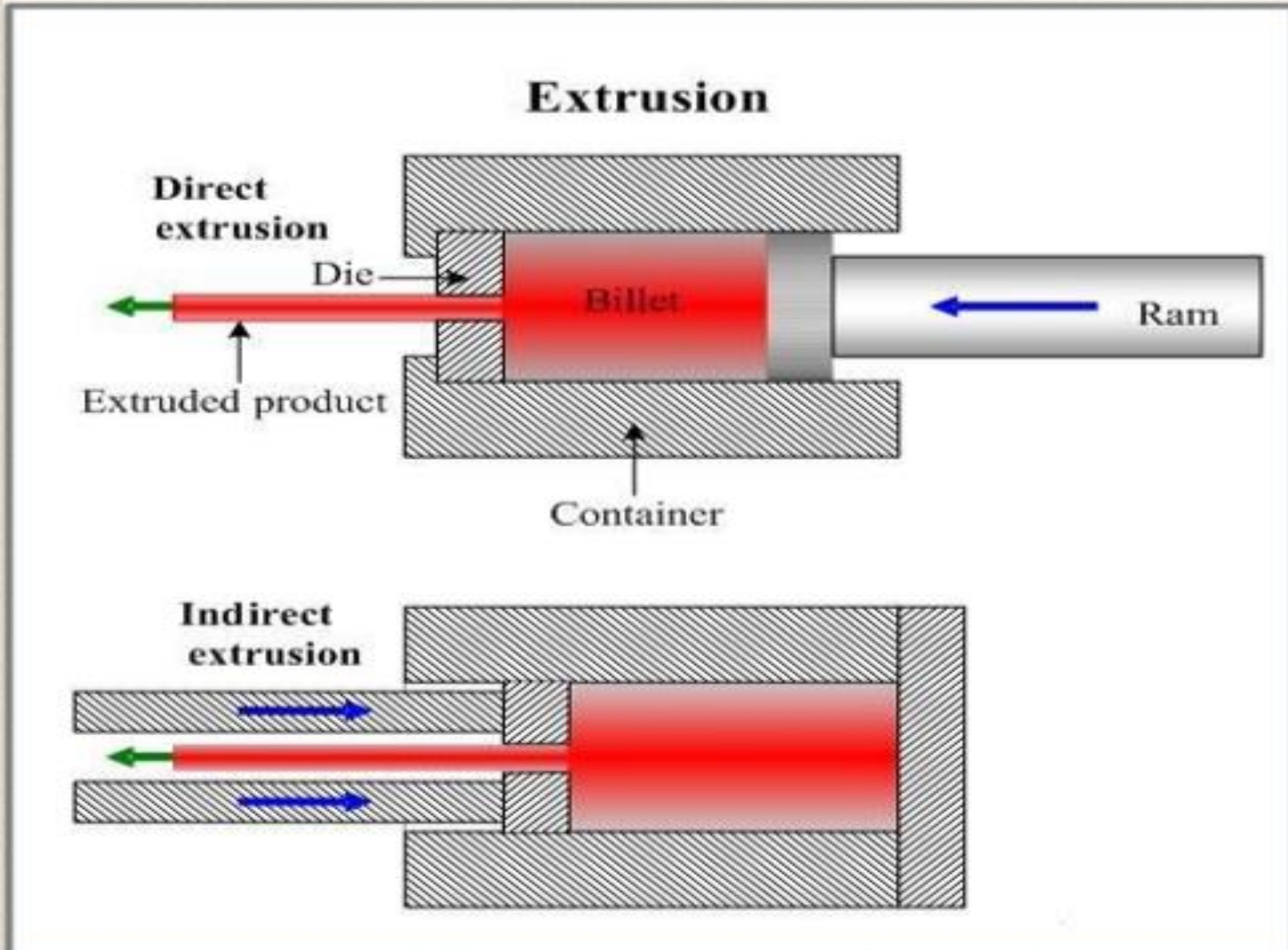


b. Indirect Extrusion

- Also called backward extrusion
- Flow of metal through the die is in the opposite direction as the movement of ram
- Hot billet(work piece) is used
- Ram used is hollow
- Billet remains stationary while die is pushed into the billet by the hollow ram
- Less force is required as compared to direct extrusion

INDIRECT EXTRUSION





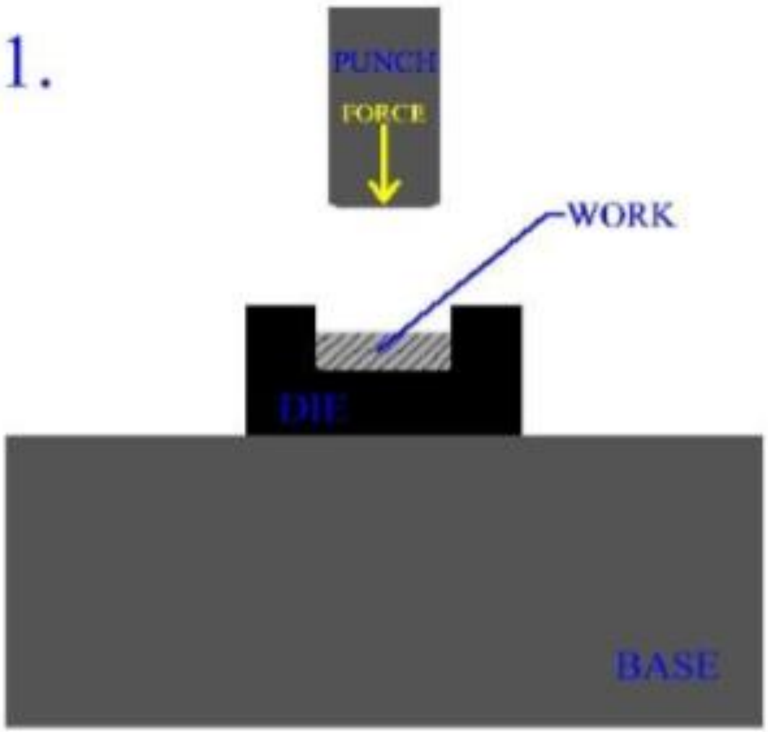
Extrusion
Direct
extrusion
Die
Billet
Ram
Extruded product
Container
Indirect
extrusion

c. Cold/Impact Extrusion

- Carried out at higher velocity
- Unheated metal is placed in the die cavity
- Punch is forced into the die cavity causing the metal to flow upwards through the gap between punch and die
- Tooth pastes tubes are made by this process

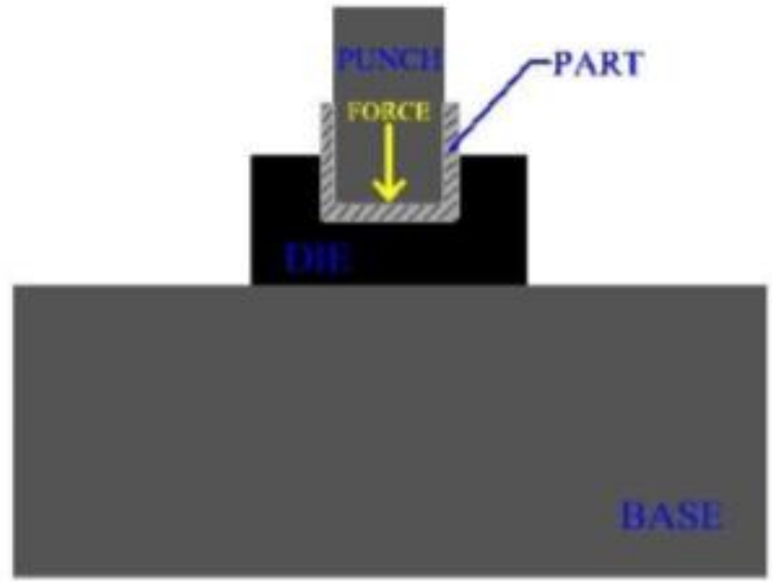
IMPACT EXTRUSION

1.



PUNCH APPROACHES WORK
AT A HIGH VELOCITY

2.



PART IS FORMED BY THE
IMPACT WITH WORK